

PAVAN KUMAR ETTA

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PROFESSIONAL SUMMARY

M.S. graduate in Computer Information Science with **3.5+** years of Software Engineering experience building backend services, distributed systems, cloud infrastructure, and storage-backed applications using Python, Go, C++, AWS, Kubernetes, SQL, Redis, and object storage. Experienced in production deployments, data reliability, fault-tolerant system design, and performance optimization.

EDUCATION

University of Massachusetts, Dartmouth <i>GPA: 4.0</i>	North Dartmouth, USA
M.S. in Computer Information Science, Data Science	Aug 2024 - May 2026
Indian Institute of Information Technology <i>GPA: 3.2</i>	SriCity, INDIA
B.Tech in Electronics and Communication Engineering	Aug 2017 - June 2021

WORK EXPERIENCE

Graduate Research Assistant University of Massachusetts, Dartmouth USA	Mar 2025 – Dec 2025
- Architected a Next.js-based web application with a Flask backend and MongoDB, enabling users to upload financial documents and view detailed reports generated through an NLP pipeline. - Integrated LangChain for AI workflow orchestration and optimized ChatGPT-based text extraction, improving extraction accuracy by 30%; implemented BERT-based classification with 93% accuracy.	
Software Engineer II Rhenix Lifesciences LLP India	Aug 2021– Aug 2024
- Built iOS, Android, and web features for the CureOrg platform, enabling patients to link 5+ wearables and view daily heart rate, sleep, steps, calories, activity, distance, and elevation metrics through Fitbit, Garmin, Withings, and Apple Health integrations. - Automated daily health-data retrieval, reducing collection effort by ~90%, by building Python ingestion jobs and Django commands on EC2 to manage authentication, refresh tokens, parse JSON, retry failed requests, and store validated records. - Improved research-data reliability by converting 5-second heart-rate signals into 1-minute windows using SQL, validating API data, handling missing records, and storing metrics in MySQL on AWS EC2. - Reduced sync debugging time by ~40% by building internal monitoring views with validation logs and AWS CloudWatch alerts to identify failed API calls, delayed jobs, and missing records faster. - Enhanced health-metrics dashboard performance by up to 70% by optimizing Django REST API responses, MySQL query patterns, indexing, pagination, and Redis caching for frequently accessed user metrics. - Accelerated backend and dashboard deployments by 80% by containerizing services with Docker and automating production deployments on AWS EKS using GitLab CI/CD pipelines.	
Software Engineer Intern Cure Science Institute USA	Jun 2021 – Aug 2021
- Developed a BLE-based asset tracking system using an Android app, BLE beacons, gateways, and REST APIs, enabling real-time location data uploads to AWS S3 with persistent background services for 24/7 tracking and improved battery efficiency. - Built an admin analytics dashboard using React, Google Maps, AWS API Gateway, DynamoDB & RDS, CloudWatch, and scheduled jobs to monitor device health, visualize coordinates, detect offline gateways, and analyze asset movement patterns.	

SKILLS

Programming: Python, JavaScript, TypeScript, C/C++, Java, Go
Backend & APIs: Django, Flask, FastAPI, REST APIs, gRPC, GraphQL, Event-Driven Architecture, System Design, Postman
Web Tools: jQuery, Angular, Node.js, Vue.js, Next.js, Flutter, Nginx, Figma, HTTPS
Databases & Caching: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch
Cloud & Infra: AWS (Lambda, EC2, S3, RDS, MSK, EKS), GCP, Azure, Terraform, Serverless
Data Engineering: Apache Spark, PySpark, Airflow, Snowflake, Hadoop, dbt
DevOps & Observability: Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI, Jenkins, Prometheus, Grafana

PROJECTS

Cloud-Native API Gateway & Rate Limiter [GITHUB]	(Python, Go, AWS, Redis, Docker, Kubernetes)
Designed a cloud-native API gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.	
Fault-Tolerant Sharded Key-Value Store [GITHUB]	(C++, Raft, SQLite)
Engineered a fault-tolerant sharded key-value store in C++ using Raft consensus and SQLite, implementing replicated state machines across multiple servers to maintain consistency, support failure recovery, and achieve 94.99% availability.	
Wandermate [GITHUB]	(React, TypeScript, Node.js, MongoDB, Socket.IO)
Developed a full-stack travel platform using React, TypeScript, Node.js, and MongoDB, integrating Socket.IO for real-time messaging, PropelAuth for secure authentication, Mapbox for interactive maps, and reusable APIs for third-party integrations.	
Trip Planner [GITHUB]	(React.js, TypeScript, Go, DynamoDB, React Native, AWS)
Designed and deployed a full-stack trip-planning platform using React, TypeScript, Go, DynamoDB, AWS Amplify, and Google Places API to generate personalized itineraries and sync them with a React Native mobile app for on-the-go access.	